Ch 8.1: Decision Trees

Lecture 24 - CMSE 381

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Mon, Nov 4, 2024

Announcements

Last time:

- Cubic Splines

This lecture:

- 8.1 Decision Trees

Announcements:

- HW #7 Sun, 11/10
- Projects

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
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33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
	Wed	11/27	Virtual: Project office hours		
	Fri	11/29	No class - Thanksgiving		
	Mon	12/2	Review		
	Wed	12/4	**Midterm #3**		
	Fri	12/6	No class - EGR Design Day		Project due

Section 1

Decision Trees

Big idea

Previously wanted

$$f(x_1, \ldots, x_p) = \hat{y}$$

Make a sequence of decisions to come to conclusion

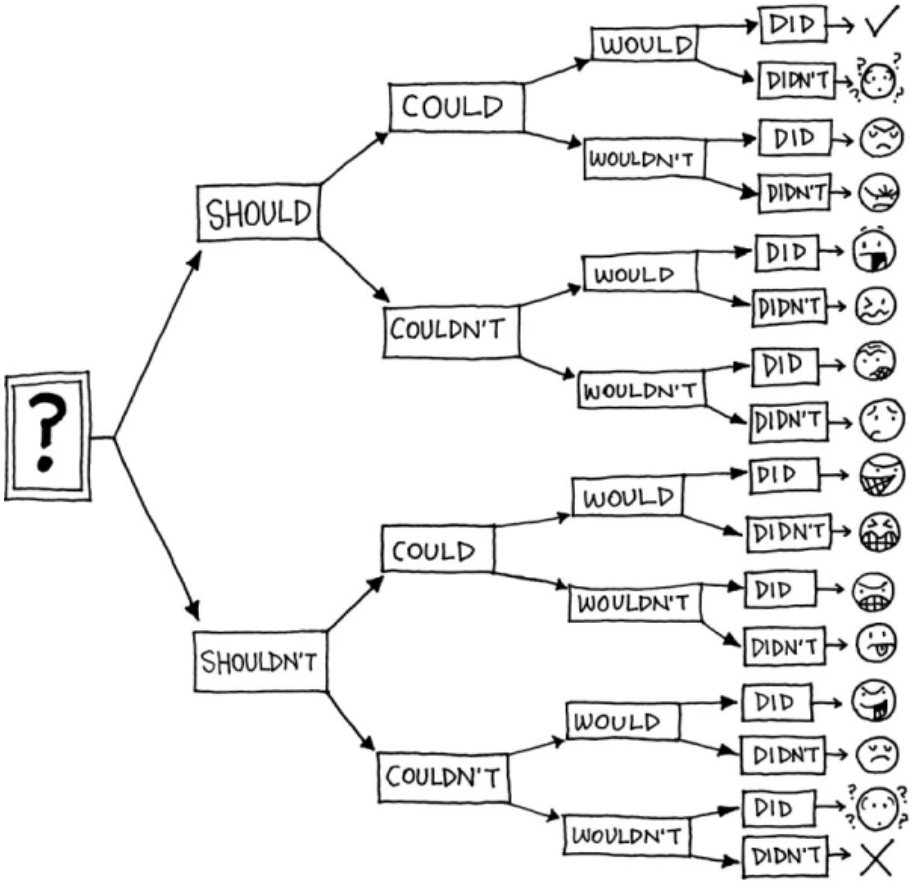

Image: https://marekbennett.com/2014/02/14/decision-tree/

Subset of Hitters data

	Hits	Years	Salary	LogSalary
1	81	14	475.0	6.163315
2	130	3	480.0	6.173786
3	141	11	500.0	6.214608
4	87	2	91.5	4.516339
5	169	11	750.0	6.620073
...	...	...	...	...
317	127	5	700.0	6.551080
318	136	12	875.0	6.774224
319	126	6	385.0	5.953243
320	144	8	960.0	6.866933
321	170	11	1000.0	6.907755

Clean up

- Removed obs w/o salary
- Predict log salary

First decision tree example

	Hits	Years	LogSalary
1	81	14	6.163315
2	130	3	6.173786
3	141	11	6.214608
4	87	2	4.516339
5	169	11	6.620073
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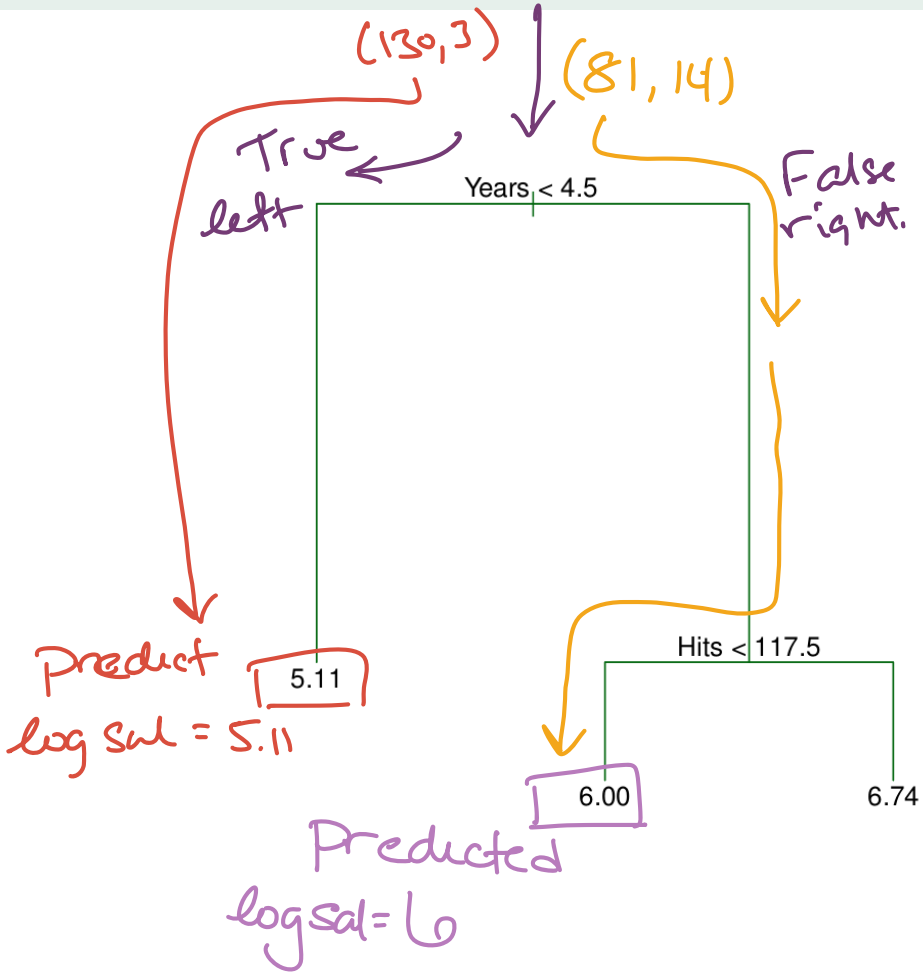
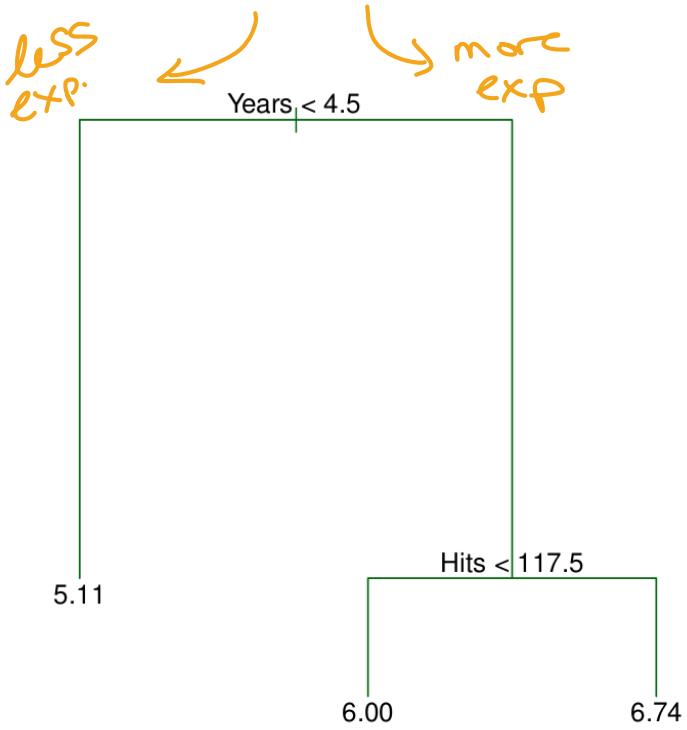


Interpretation of example



1st decision based on years
↳ most important factor for predicting salary

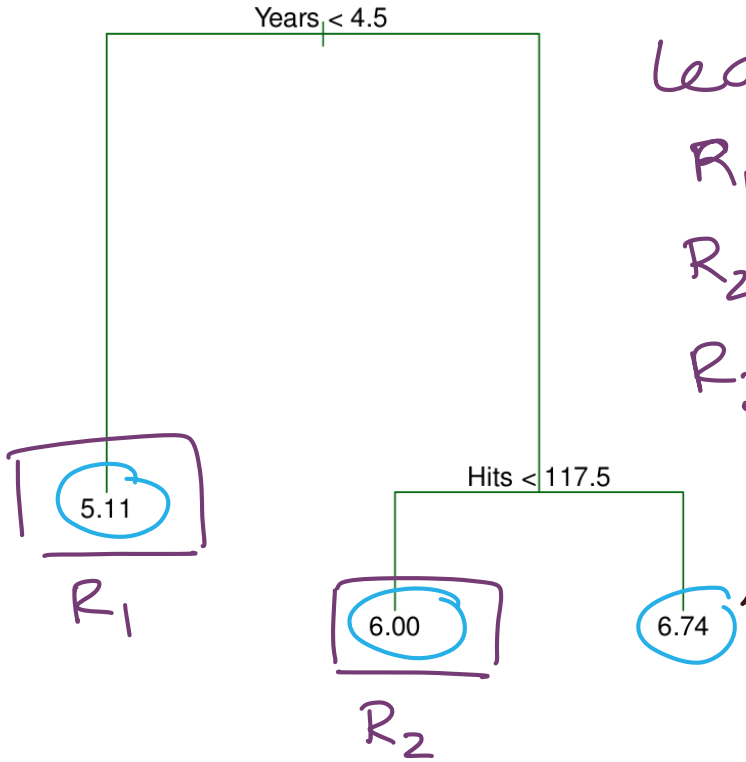
Players w/ less experience earn lower salaries than more exp.
5.11 6.00 or 6.74

less exp players not affected by # hits, but matters for more exp. players.

Likely oversimplification, but simple = easier interpretation

Coding a regression decision tree

Regions defined by the tree



Tree "upside down"

leaves correspond to regions of input space

$R_1 = \{(\text{hits}, \text{years}) \mid \text{years} < 4.5\}$

$R_2 = \{(\text{hits}, \text{years}) \mid \text{years} \geq 4.5, \text{hits} < 117.5\}$

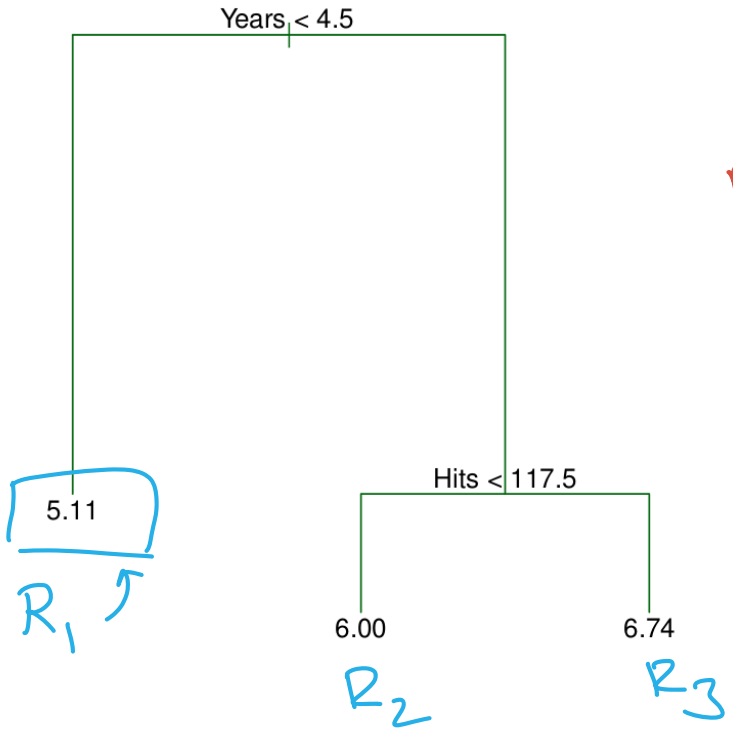
$R_3 = \{(\text{hits}, \text{years}) \mid \text{years} \geq 4.5, \text{hits} \geq 117.5\}$

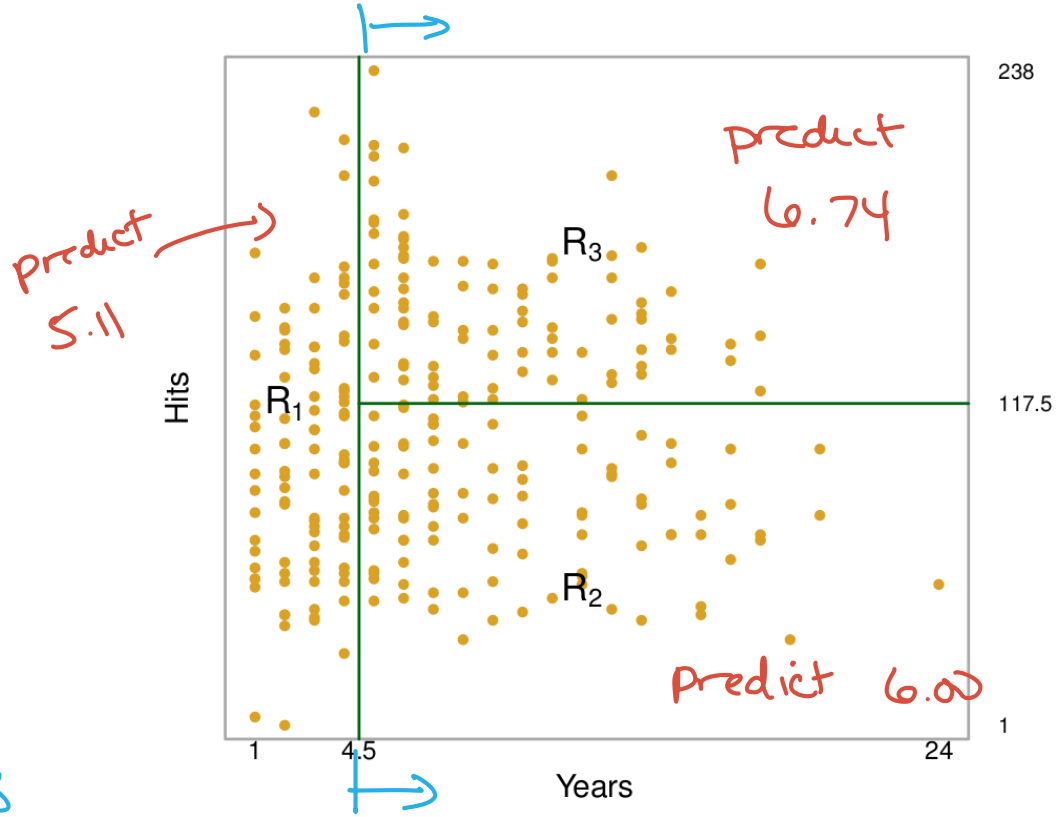
leaves no more decisions available

other splits called "internal nodes"

connections are called "branches" or "edges"

Viewing Regions Defined by Tree

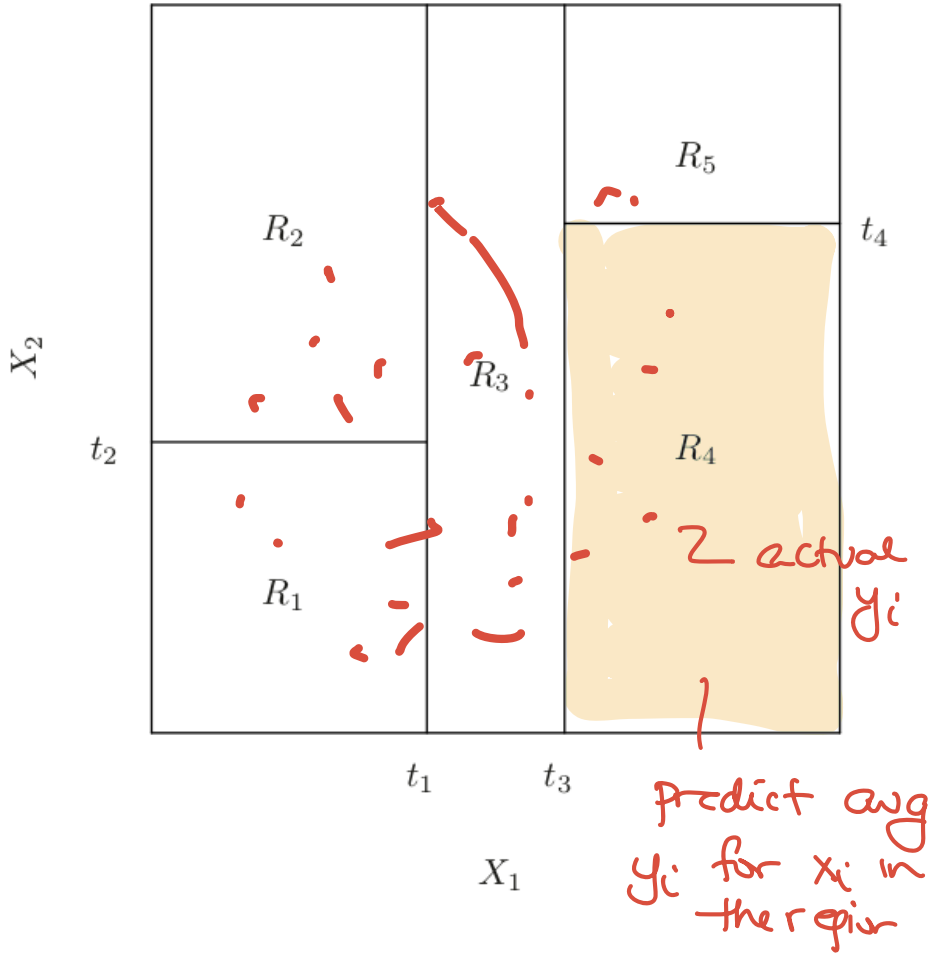




How do we actually get the tree? Two steps

1. We divide the predictor space – that is, the set of possible values for $X_1, X_2, \cdots, X_p$ — into J distinct and non-overlapping regions, $R_1, R_2, \cdots, R_J$.
2. For every observation that falls into the region R_j, we make the same prediction = the mean of the response values for the training observations in R_j.

y



Step 1: How do we decide on R_js?

Can't check all possible $R_1, \ldots, R_J$ combos.

Recursive binary splitting

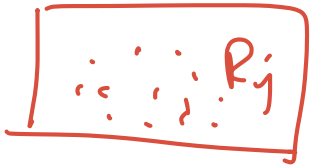
Goal:

Find boxes $R_1, \cdots, R_J$ that minimize

$$RSS = \sum_{j=1}^{J} \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

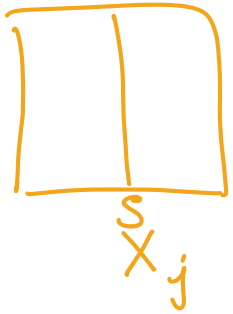
data x_i in the region R_j

$\hat{y}_{R_j}$ = mean response for training observations in jth box



Recursive Binary Splitting

- Pick X_j $X_1, \ldots, X_p$
- Pick s so that splitting into $\{X \mid X_j < s\}$ and $\{X \mid X_j \geq s\}$ results in largest possible reduction in RSS

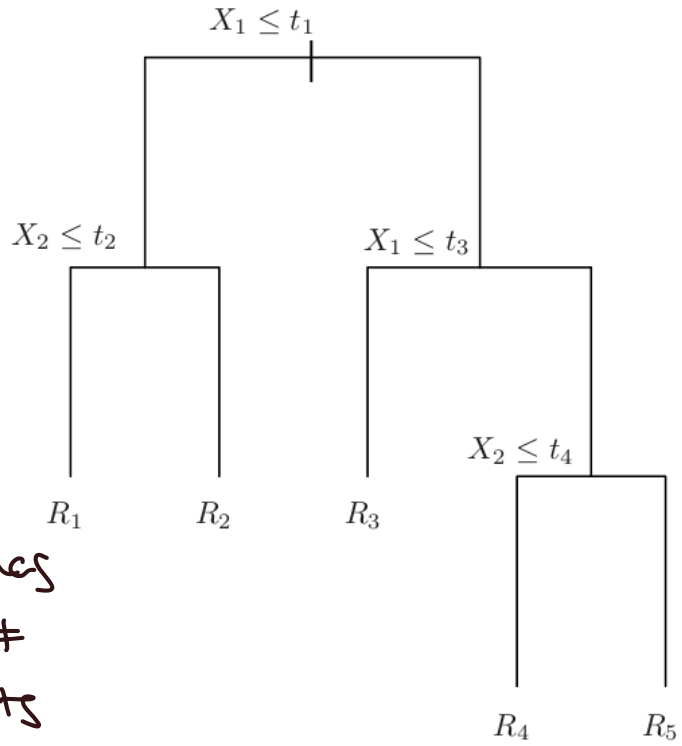


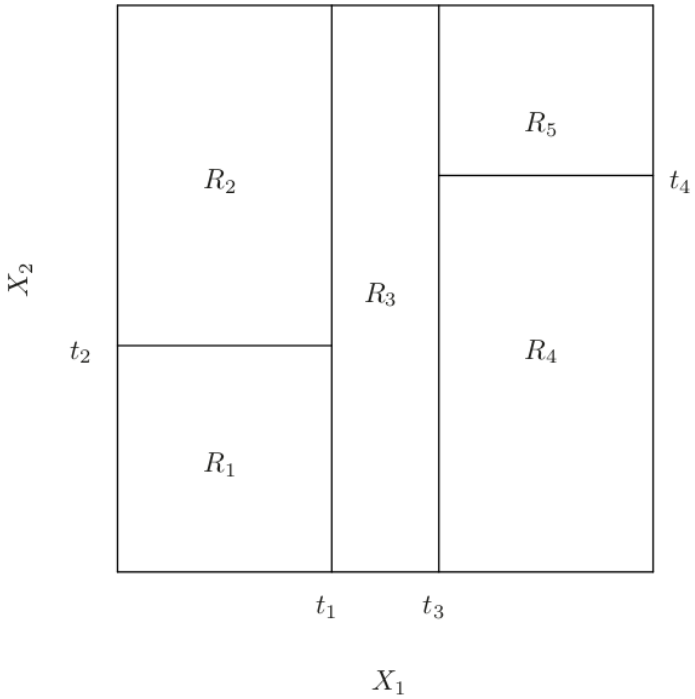
$$R_1(j,s) = \{X \mid X_j < s\}$$
$$R_2(j,s) = \{X \mid X_j \geq s\}$$

$$\underbrace{\sum_{i \mid x_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2}_{\text{now in left split}} + \underbrace{\sum_{i \mid x_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2}_{\text{now in the right split}}$$

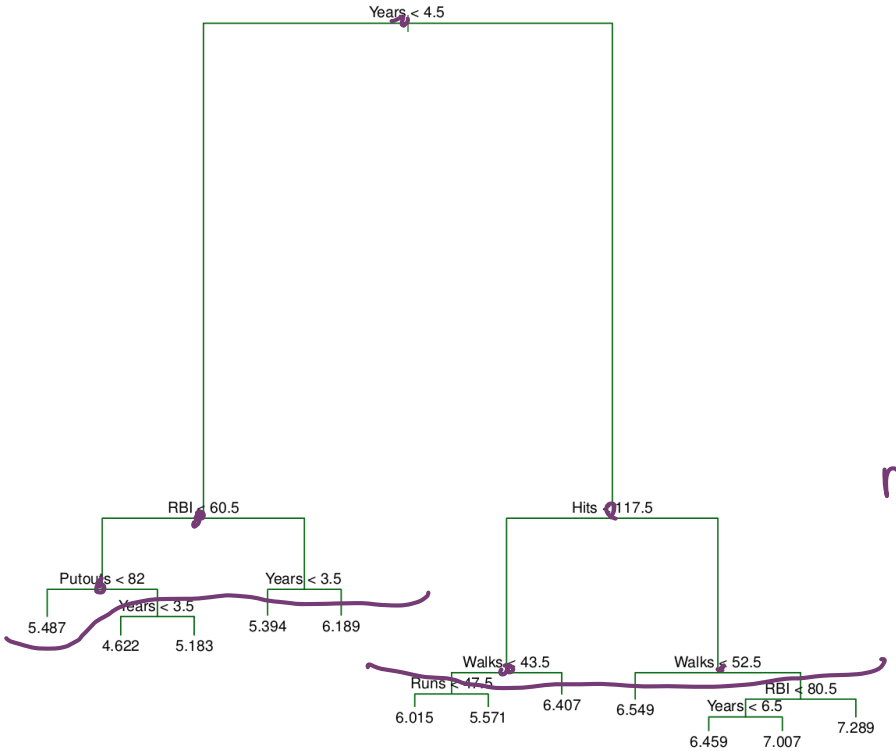
Rinse and repeat

- Split repeatedly
- Each step picks some region and splits in "half"
- Stopping criterion
  - ↳ each region has < some max # of data pts





Pruning



- Big trees ⇒ potential for overfitting

Plan A: pick some max depth

"How many steps until we reach a leaf"

max depth 3

Weakest Link Pruning

Also called Cost complexity pruning

For every α, there is a subtree T that minimizes:

$$\sum_{m=1}^{|T|} \sum_{i|x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$$

- $|T|$ = number of terminal nodes of T
- R_m is rectangle for mth terminal node
- $\hat{y}_{R_m}$ is mean of training observations in R_m

Algorithm version

Algorithm 8.1 *Building a Regression Tree*

1. Use recursive binary splitting to grow a large tree on the training data, stopping only when each terminal node has fewer than some minimum number of observations.

2. Apply cost complexity pruning to the large tree in order to obtain a sequence of best subtrees, as a function of α.

3. Use K-fold cross-validation to choose α. That is, divide the training observations into K folds. For each $k = 1, \ldots, K$:

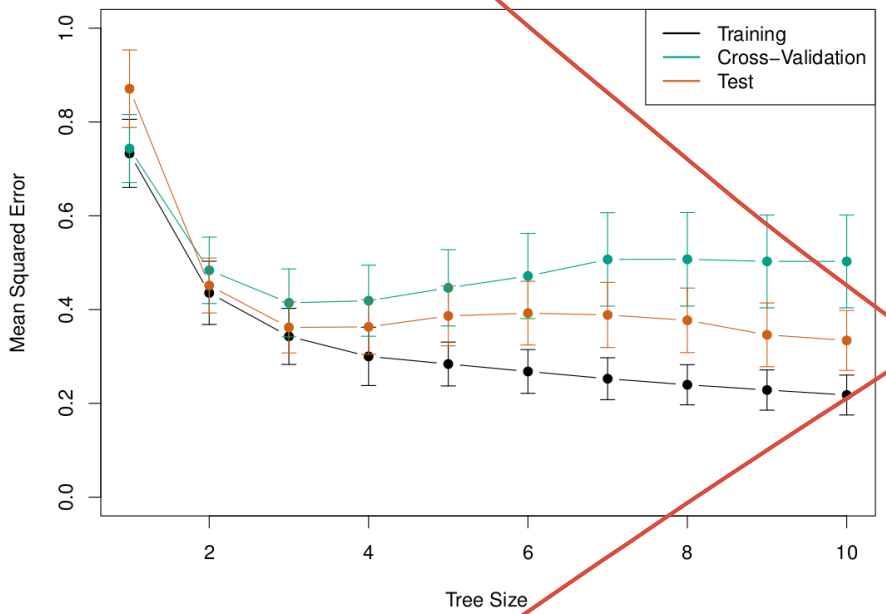
   (a) Repeat Steps 1 and 2 on all but the kth fold of the training data.

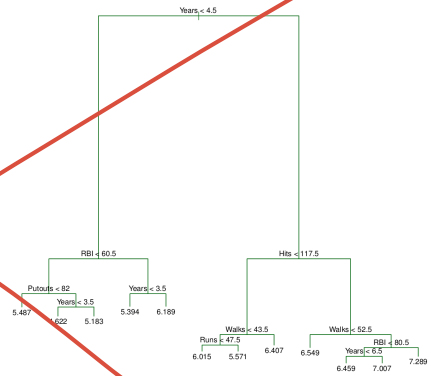
   (b) Evaluate the mean squared prediction error on the data in the left-out kth fold, as a function of α.

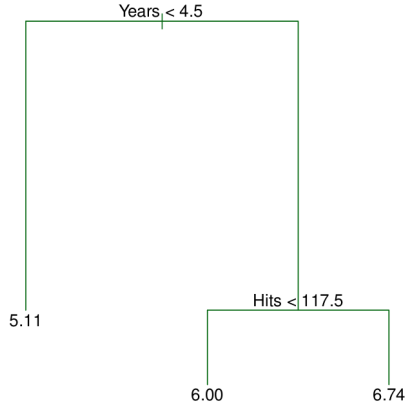
   Average the results for each value of α, and pick α to minimize the average error.

4. Return the subtree from Step 2 that corresponds to the chosen value of α.

Messing with α





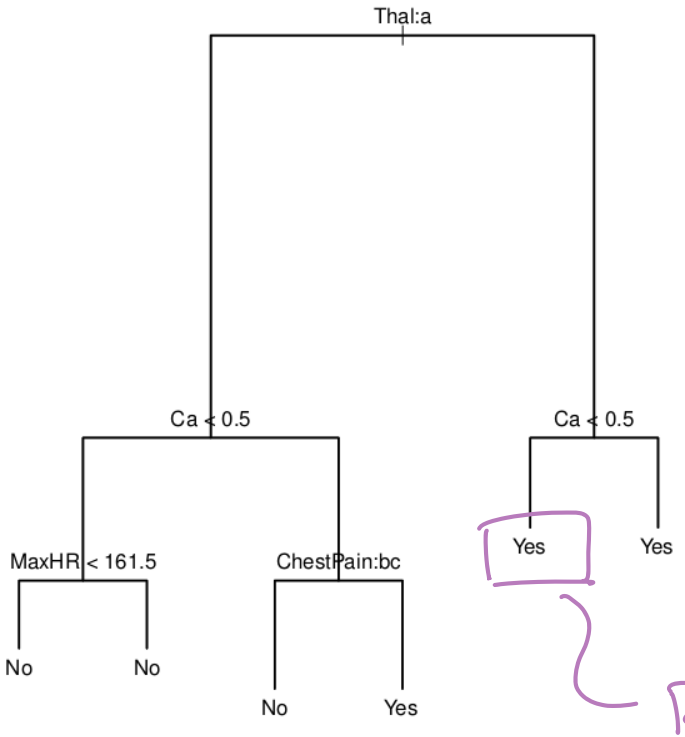


Section 2

Classification Decision Tree

Basic idea

ex from heart data



- $\hat{p}_{mk}$ = proportion of training observations in R_m from the kth class
- $E = 1 - \max_k(\hat{p}_{mk})$

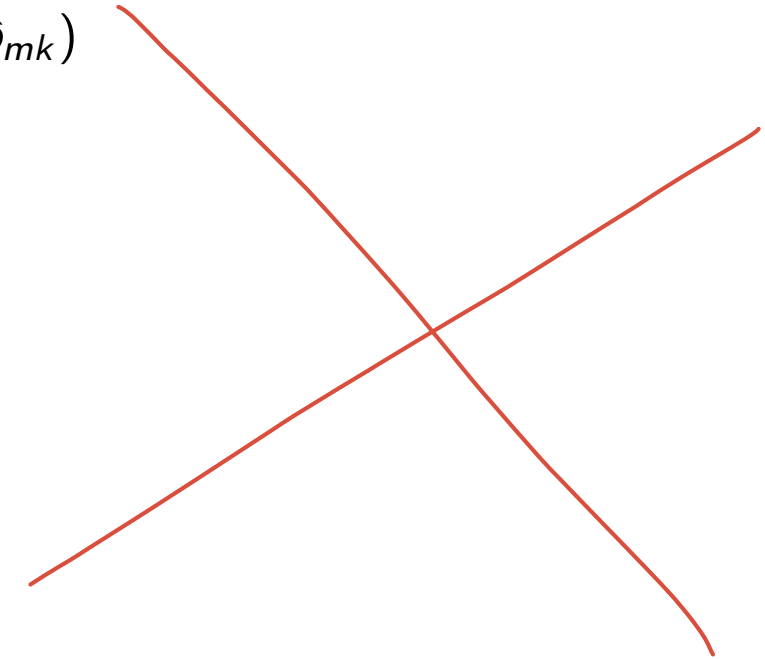
Frac of training obs incorrectly labeled

R_j, how many train samples w/ label p_1, p_2, p_3 .... p_k

Predict label w/ most training samples in R_j

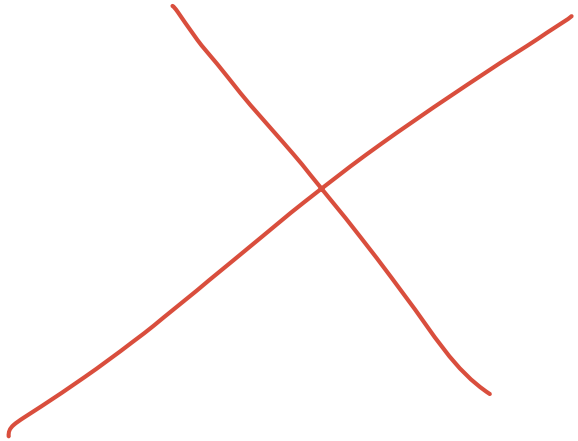
Gini index

$$G = \sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk})$$

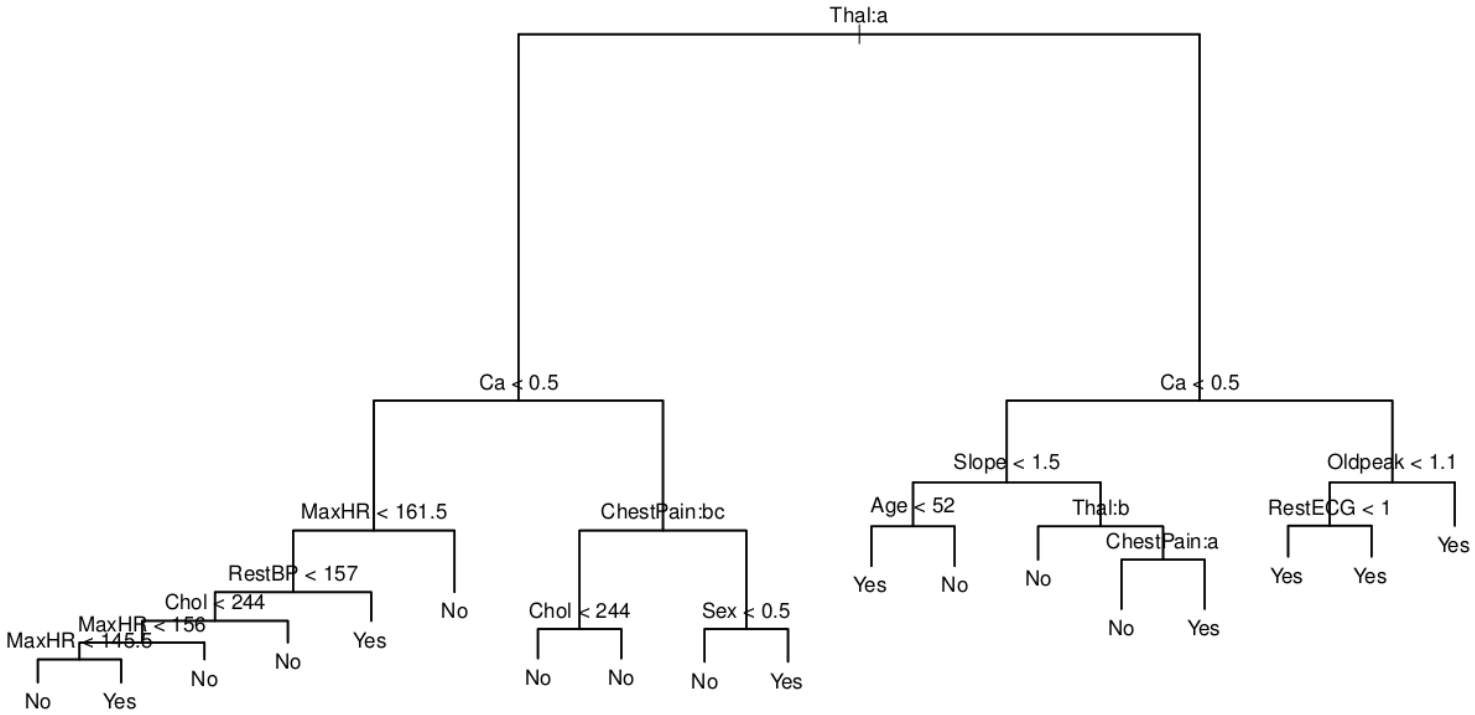


Entropy

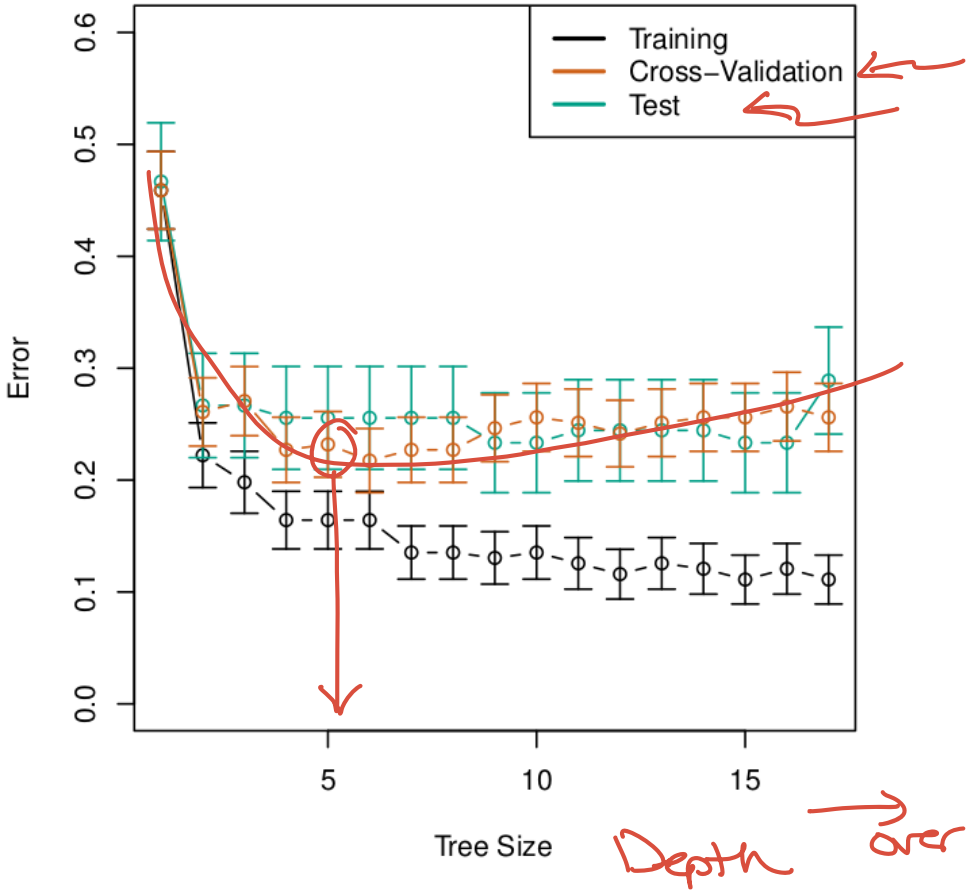
$$D = -\sum_{k=1}^{K} \hat{p}_{mk} \log \hat{p}_{mk}$$

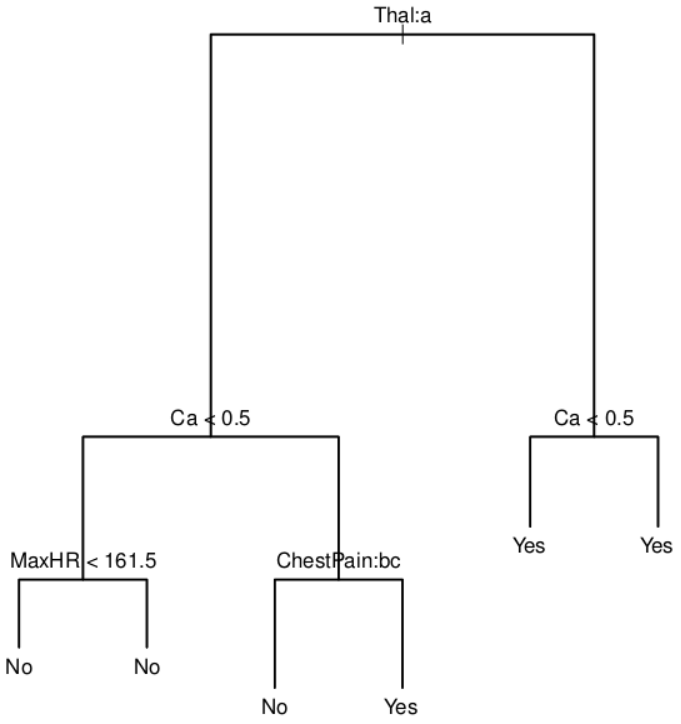


Example


Thal:a
Ca < 0.5
Ca < 0.5
MaxHR < 161.5
ChestPain:bc
RestBP < 157
Chol < 244
MaxHR < 156
MaxHR < 145.5
No
Yes
No
No
Yes
No
Chol < 244
Sex < 0.5
No
No
No
Yes
Slope < 1.5
Oldpeak < 1.1
Age < 52
Thal:b
RestECG < 1
Yes
Yes
Yes
ChestPain:a
Yes
No
No
No
Yes

Pruning the example





More coding!

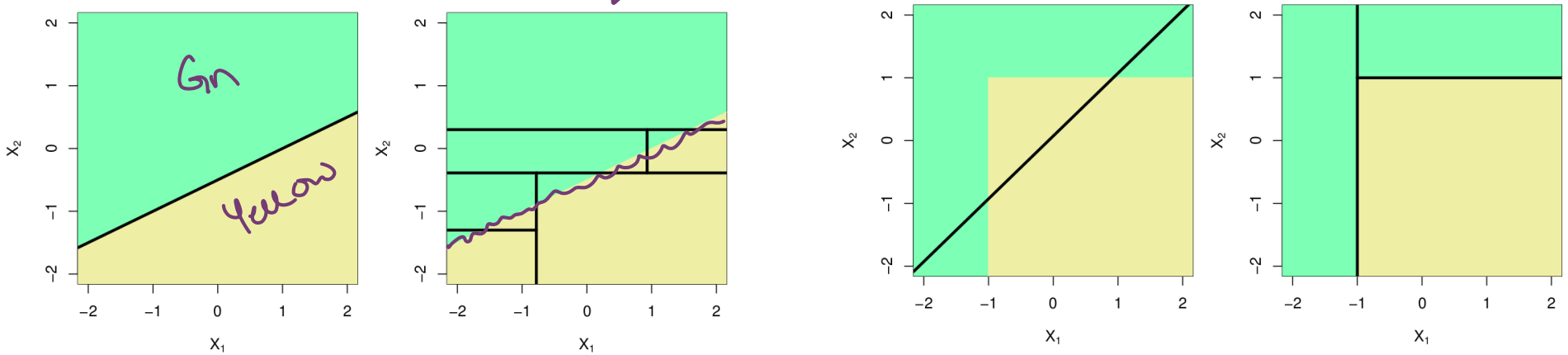
Linear models vs trees

actual from linear

decision tree isn't doing great

linear does less well

trees does better



Pros/Cons

Pros:

Easiest to explain.

Trees can be displayed graphically
easy to interpret, esp. if small

Easy to use categorical predictors!
No dummy variables needed!

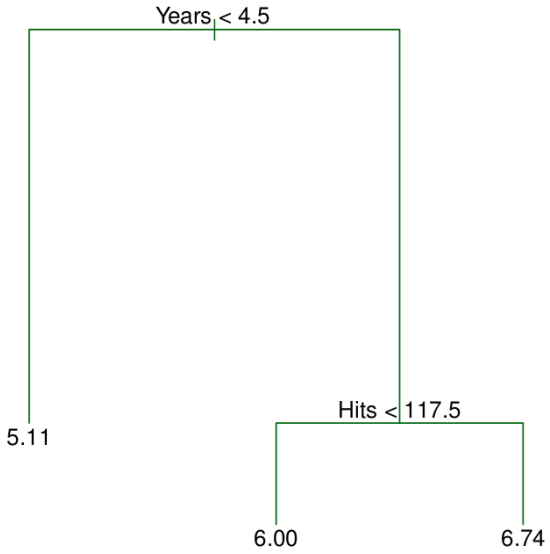
Cons:

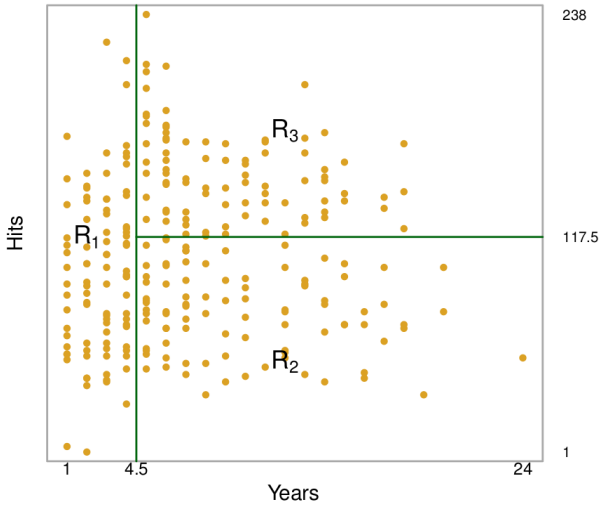
Not as accurate as
other methods of
classification/regression

Not robust
↳ small changes
in input data
can result in
very diff tree!

TL:DR

- Split into regions by greedily decreasing RSS
- Prune tree by using cost complexity
- Not robust - Next time, figure out how to aggregate trees





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